Topic 1[Method/Function Overloading]

Problem Statement: Define a class Test where overload a method Sum() to sum numbers sent from main() function.

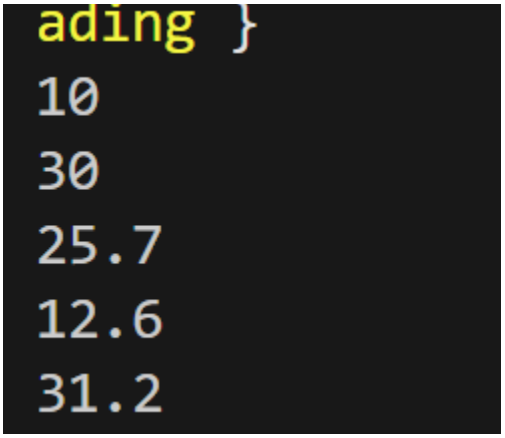
Solution:

```
#include<bits/stdc++.h>
using namespace std;
class Test
{
    public:
    int Sum(int t)
    {
        return t;
    }
    int Sum(int a, int b)
    {
        return a+b;
    }
    double Sum(double a, int b)
    {
        return a+b;
    }
    double Sum(int a, double b)
    {
        return a+b;
    }
    double Sum(double a, double b)
    {
        return a+b;
    }

};
int main()
```

```
{
    Test t;
    cout << t.Sum(10) << endl;
    cout << t.Sum(10,20) << endl;
    cout << t.Sum(5.7,20) << endl;
    cout << t.Sum(10,2.6) << endl;
    cout << t.Sum(10.5,20.7) << endl;
    return 0;
}
```

Output:



Topic 2 [Binary Operator Overloading]: Suppose in a AC circuit, there are 3 impedances z1=3+j4, z2=4-j3 and z3=j6 are connected in parallel. Now find the current in the circuit if input voltage is 100+j50. Implement operator overloading concept for your calculation. Use class Circuit and initialize the impedance values (real & img) by a constructor.

Solution:

```
#include <iostream>
using namespace std;

class Circuit {
private:
```

```
    double real;
    double img;

public:
    Circuit(double r = 0, double i = 0) {
        real = r;
        img = i;
    }
    Circuit operator+(Circuit c) {
        return Circuit(real + c.real, img + c.img);
    }
    Circuit operator/(Circuit c) {
        double denom = c.real * c.real + c.img * c.img;
        double r = (real * c.real + img * c.img) / denom;
        double i = (img * c.real - real * c.img) / denom;
        return Circuit(r, i);
    }
    void display() {
        if (img >= 0)
            cout << real << " + j" << img << endl;
        else
            cout << real << " - j" << -img << endl;
    }
};

int main() {
    Circuit z1(3, 4);
    Circuit z2(4, -3);
    Circuit z3(0, 6);
```

```
    Circuit V(100, 50);
    Circuit one(1, 0);
    Circuit y1 = one / z1;
    Circuit y2 = one / z2;
    Circuit y3 = one / z3;

    Circuit Yeq = y1 + y2 +
y3;

    Circuit Zeq = one / Yeq;

    Circuit I = V / Zeq;

    cout << "Equivalent Impedance Z = ";
    Zeq.display();

    cout << "Circuit Current I = ";
    I.display();

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\week 4> cd "d:\Satu\1
g++ binaryoprtroverloading.cpp -o binaryoprtr
yoprtroverloading }
Equivalent Impedance Z = 2.31193 + j1.70642
Circuit Current I = 38.3333 - j6.66667
```

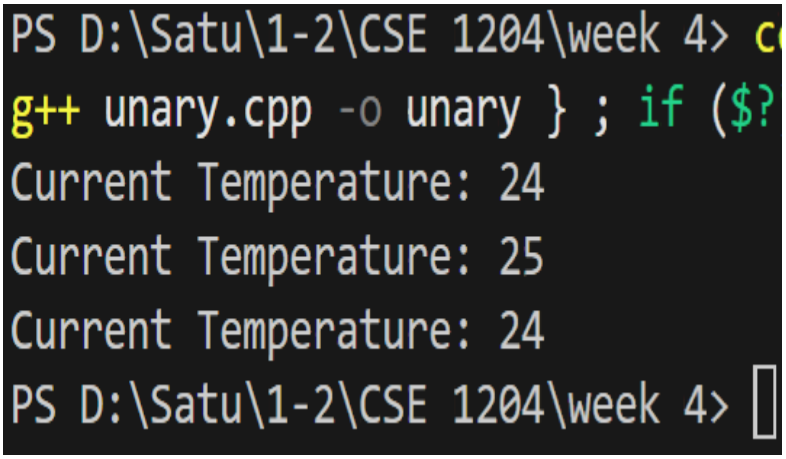
Topic 3 [Unary Operator Overload]

Problem statement: You need to write a program for remote controller of Air-conditioner. You need the control two buttons like temperature up and temperature down using the concept of unary operator overloading. The main() function would be look like:

Solution:

```
#include <iostream>
```

```
using namespace std;
class AirConditioner
{
    int temperature;
public:
    AirConditioner(int t)
    {
        temperature = t;
    }
    void operator++()
    {
        temperature++;
    }
    void operator--()
    {
        temperature--;
    }
    void display()
    {
        cout << "Current Temperature: " << temperature << endl;
    }
};
int main()
{
    AirConditioner ac(24);
    ac.display();
    ++ac;
    ac.display();
    --ac;
    ac.display();
    return 0;
}
```



Topic 4[Method/Function Overriding]

Problem statement: Write a class A with a method Print() and a derived class B with method Print() overloaded. Now observe the output when following statements are written in the main() function-

Solution:

```
#include <iostream>
using namespace std;
class A{
public:
    void Print(){
        cout << "Inside Print() of class A" << endl;
    }
};
class B : public A{
public:
    void Print(){
        cout << "Inside Print() of class B" << endl;
    }
};
int main()
{
    A a;
    a.Print();
    B b;
    b.Print();
    A a2;
    A *p1;
    p1 = &a2;
    p1->Print();
    B b2;
    A *p2;
    p2 = &b2;
    p2->Print();
    return 0;
```

```
PS D:\Satu\1-2\CSE 1204\we
g++ functionoveriding.cpp
ng }
Inside Print() of class A
Inside Print() of class B
Inside Print() of class A
Inside Print() of class A
```

```
}
```

Topic 5 [Friend Function]

Problem Statement: Using the following class, write three friend functions

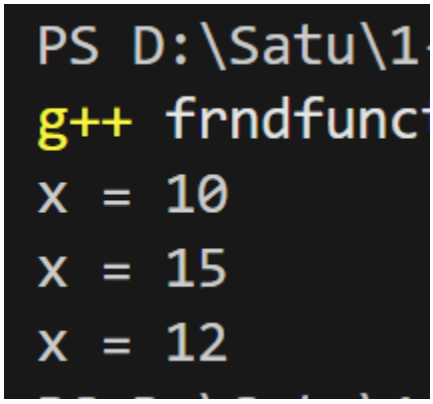
i) Add(): Assign value to the data member x

ii) IncX() : Increase the value of x by m

iii) DecX() : Decrease the value of x by n

Solution:

```
#include <iostream>
using namespace std;
class Sample
{
    int x;
public:
    Sample()
    {
        x = 0;
    }
    friend void Add(Sample &s, int value);
    friend void IncX(Sample &s, int m);
    friend void DecX(Sample &s, int n);
    void display()
    {
        cout << "Value of x: " << x << endl;
    }
};
void Add(Sample &s, int value)
{
    s.x = value;
```



```
}

void IncX(Sample &s, int m)
{
    s.x = s.x + m;
}
void DecX(Sample &s, int n)
{
    s.x = s.x - n;
}
int main()
{
    Sample obj;
    Add(obj, 10);
    obj.display();
    IncX(obj, 5);
    obj.display();
    DecX(obj, 3);
    obj.display();
    return 0;
}
```

Topic 6[Pure Virtual Function]

Problem statement: Modify the class defined in Topic 3 executes the following

```
#include <iostream>
using namespace std;

class A{
public:
```

```
    virtual void Print() = 0;
};

class B : public A{
public:
    void Print(){
        cout << "Inside Print() of class B" << endl;
    }
};
```